

DATA ANALYSIS INTERNSHIP

DAY 19 REPORT

Topic: Power BI Query Editor – Rows, Columns, Split Column, Group By, Pivot, Parse, and Fill Functions

Introduction

Power BI Query Editor is an important tool used for data cleaning, transformation, and preparation before creating dashboards and reports. It helps organize raw datasets into structured formats for accurate analysis and visualization. In this task, various Query Editor operations such as row operations, column transformations, split column methods, Group By, Pivot Column, Parse, and Fill functions were learned and implemented.

Topics Covered

1. Row Operations

Different row operations were performed in Power Query Editor, including:

- Remove Rows
- Remove Top Rows
- Remove Bottom Rows
- Remove Blank Rows
- Remove Alternate Rows
- Keep Rows

These operations help remove unwanted or duplicate data from the dataset.

2. Column Operations

Various column transformations were explored, such as:

- Rename Column
- Remove Column
- Duplicate Column

- Change Data Type
- Move Columns
- Merge Columns

These operations help organize and structure data properly for analysis.

3. Split Column

The Split Column feature was used to divide combined data into multiple columns using:

- Delimiter (comma, space, hyphen, etc.)
- Number of Characters
- Positions
- Uppercase to Lowercase Transition
- Digit to Non-Digit Transition

This feature helps separate names, IDs, dates, and product codes into structured columns.

4. Group By Function

The Group By option was used to summarize and aggregate data.

Functions used:

- Sum
- Count
- Average
- Minimum
- Maximum

Example

Revenue was grouped by:

- Product
- Region
- Category

This helps create summarized business reports and insights.

5. Pivot Column

The Pivot Column feature was used to convert row values into separate columns.

Example

Monthly sales values stored in rows were converted into separate month columns for easier comparison and reporting.

This feature helps:

- Create summary tables
 - Improve data readability
 - Simplify analysis
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6. Parse Function

The Parse function was used to convert raw or unstructured data into a readable and usable format.

Examples:

- Parsing date values into proper date format
- Converting imported data into structured columns

This helps prepare datasets accurately for reporting and visualization.

7. Fill Function

The Fill function was used to replace blank or null values automatically.

Types used:

- Fill Down
- Fill Up

This operation helps maintain consistent data and remove empty values from datasets.

Tools Used

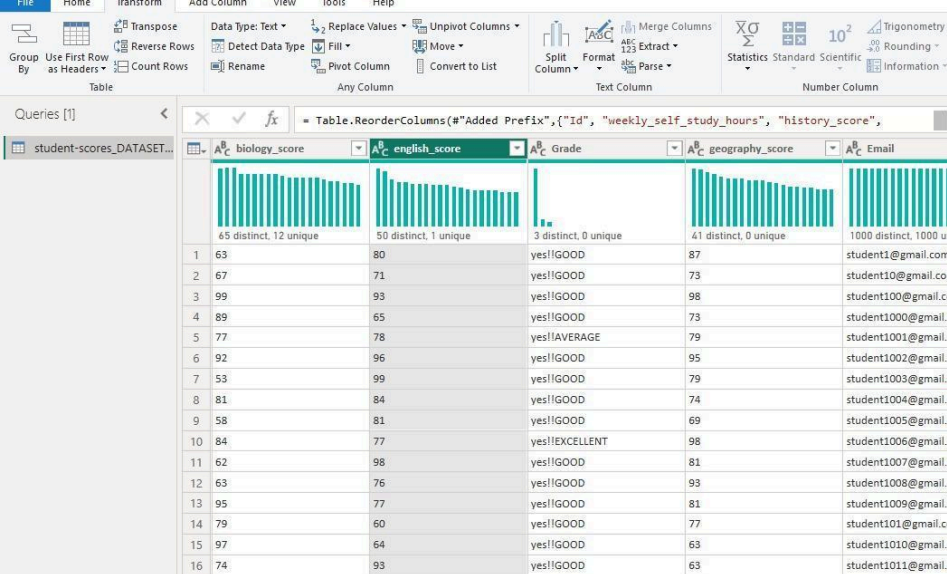
- Microsoft Power BI
- Power Query Editor

Outcome

The Query Editor operations were successfully implemented for data cleaning, transformation, formatting, and summarization. These techniques improved data organization and prepared the dataset effectively for dashboards and reports.

Conclusion

Through this task, I learned how to use Power BI Query Editor for row and column operations, split column methods, Group By functions, Pivot Columns, Parse functions, and Fill operations. I also improved my understanding of data transformation and preparation techniques in Power BI.



The screenshot displays the Power BI Query Editor interface. The top ribbon includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The Transform tab is active, showing various data manipulation options like Transpose, Reverse Rows, Count Rows, Detect Data Type, Fill, Pivot Column, Convert to List, Split Column, Format, Merge Columns, Extract, Parse, Statistics, Standard, Scientific, Rounding, and Information. The main area shows a table named 'student-scores_DATASET...' with the following columns: biology_score, english_score, Grade, geography_score, and Email. Each column has a small bar chart above it indicating the distribution of data. The table contains 16 rows of data, with the first row being the header row. The formula bar at the top shows the query: 'Table.ReorderColumns(#"Added Prefix",{"Id", "weekly_self_study_hours", "history_score", ...)'.

	biology_score	english_score	Grade	geography_score	Email
1	63	80	yes!!GOOD	87	student1@gmail.com
2	67	71	yes!!GOOD	73	student10@gmail.com
3	99	93	yes!!GOOD	98	student100@gmail.com
4	89	65	yes!!GOOD	73	student1000@gmail.com
5	77	78	yes!!AVERAGE	79	student1001@gmail.com
6	92	96	yes!!GOOD	95	student1002@gmail.com
7	53	99	yes!!GOOD	79	student1003@gmail.com
8	81	84	yes!!GOOD	74	student1004@gmail.com
9	58	81	yes!!GOOD	69	student1005@gmail.com
10	84	77	yes!!EXCELLENT	98	student1006@gmail.com
11	62	98	yes!!GOOD	81	student1007@gmail.com
12	63	76	yes!!GOOD	93	student1008@gmail.com
13	95	77	yes!!GOOD	81	student1009@gmail.com
14	79	60	yes!!GOOD	77	student101@gmail.com
15	97	64	yes!!GOOD	63	student1010@gmail.com
16	74	93	yes!!GOOD	63	student1011@gmail.com